

RESEARCH ARTICLE

A ~~real time~~ real-time reservoir inflow forecast evaluation framework

Cameron Bracken | Youngjun Son | Vince Tidwell | Nathalie Voisin

¹Pacific Northwest National Laboratory,
Washington, USA

Correspondence

Corresponding author Cameron Bracken,
Email: cameron.bracken@pnnl.gov

Abstract

Recent advances in data-driven New hydrologic forecasts have led to a multitude of new and rapidly improving streamflow forecast products and technologies. These often lead give hydropower producers operators more options, but operators still need to question determine whether such advances translate into increases in revenue and whether implementing these new forecasts a product improves decisions enough to justify the additional associated costs its cost. To address this, we We present a general framework that can be used in real time (or near real time) procedure for evaluating competing reservoir inflow forecast evaluation forecasts as they are used. The framework calls for an iterative approach which involves direct operator feedback in Operators help select and revise the metrics, and the evaluation processes the same data available in operations. The evaluation can include forecast Standard verification and validation, but also includes other qualitative or specialized metrics of forecast quality are paired with measures tied to site-specific decisions. We also present a successful application of this evaluation framework apply the procedure to three reservoirs in the Great River Hydro system on the Connecticut River in the Northeast U.S. While there is no single evaluation approach that can apply to all hydrologic forecasts, we The commercial product labeled B generally matched the in-house forecast and outperformed product A. A cumulative-error stress test also present showed that B had a range-lower probability of options of both traditional exceeding the forebay compliance band during the first day at all three dams. By days two and tailored metrics that will be suitable three, exceedance probabilities were high for a diverse set both commercial products, showing the horizon at which passive use of situations and systems either forecast would be untenable. The goal of this framework is to provide hydropower operators with information necessary for evaluating how much a particular evaluates forecast product will improve their operations and ultimately inform ongoing investment decisions quality rather than economic value; a value assessment requires a scheduling model, such as whether to purchase external forecast products or improve existing internal forecasting capabilities operating constraints, and market prices.

KEY WORDS

hydropower, streamflow forecasting, real-time evaluation, verification, validation

1 | INTRODUCTION

~~Changing environmental regulation, diversifying and increasing~~ Environmental regulation, changing water demands, and ~~growing operational uncertainties are requiring more precise operations of~~ operational uncertainty require reservoir and

hydropower ~~systems operators to make more precise decisions~~ (e.g., Lee et al. (2025) and Guo and Liu (2024) and Badr et al. (2023) and Aljoda and Jain (2021)). This ~~desire for increased precision has driven interest in improved~~ has increased interest in streamflow forecasting tools ~~(Yang et al. 2021 and Pagano et al. 2014 and Liu et al. 2012)~~. While ~~numerous commercial and in-house options are available, water managers are faced with the challenge of evaluating whether more sophisticated forecasting tools will actually improve their operations and justify the associated financial investment~~(Yang et al. 2021 and Pagano et al. 2014 and Liu et al. 2012). The practical question is whether a new product improves operations enough to justify its cost.

~~The quality of hydrologic forecasts~~ Hydrologic forecasting and hydropower scheduling ~~has rapidly improved in recent years due to the emergence of new data driven approaches (Kratzert et al. 2019)~~ have advanced through data driven models (Kratzert et al. 2019) and scheduling optimization (Zhang et al. 2024). ~~Despite these improvements verification remains a challenge and traditional verification approaches may not be applicable under new operational paradigms. This study seeks to provide insight into the evaluation of contemporary forecast models and guidance on how to assess their value.~~ Forecast verification still commonly relies on hindcasts and standard statistical metrics over long records. Hindcasts are often unavailable because they are costly to produce or require data that were not archived. Archived operational forecasts provide another option, but they do not capture products that change during the evaluation or forecasts adjusted by operators. We address the case in which a reservoir operator must compare products under current operating conditions without a hindcast.

~~Pairing a forecasting tool with the needs of a particular hydropower operator is complicated by a number of factors. The quantity and quality of data concerning meteorological (precipitation, snow pack) and hydrological (reservoir storage, soil moisture) conditions within the watershed vary locationally as do the efficacy of meteorological forecasting (Hay et al. 2023; Chen et al. 2021; Wood et al. 2016). Watershed characteristics vary as does the infrastructure used to manage basin resources (Pham et al. 2021). Operators are often subject to very different management regimes (e.g.,~~ No metric set fits every system. Meteorological and hydrologic data vary among basins, as do watershed properties and infrastructure (Pham et al. 2021). Operating rules range from run-of-river , rule curves) (Zajac et al. 2017). ~~Management objectives also vary according to forecast lead time as well as normal versus extreme event conditions. Additionally, there are decisions concerning choice of model (conceptual, physical, empirical, AI enhanced) as well as its calibration and parameterization~~ constraints to seasonal rule curves (Zajac et al. 2017). Decisions also change with lead time and flow regime. An evaluation must therefore connect statistical performance to the decisions made at a particular site.

~~To help provide some clarity concerning the performance of various forecasting tools under differing informational, physical, and operational conditions, a range of head-to-head evaluation exercises have been performed. HEPEx (which stands for Hydrologic Ensemble Prediction EXperiment), which “seeks to advance the science and practice of hydrological ensemble prediction and its use in impact and risk-based decision making,” has organized evaluations in the context of~~ Several programs

~~compare forecasts under common conditions. HEPEx has organized~~ community experiments and testbeds^(†). ~~The~~, and the U.S. Bureau of Reclamation has ~~also performed a series of forecast rodeos in a prize competition environment~~ run forecast rodeos[‡]. Real time forecast evaluations such as this (as opposed to retrospective evaluations) tend to be favored by the industry ~~due their ability to fit in with existing operations (Krajewski et al. 2021). While such exercises provide valuable insight into the complexities and potential solutions for streamflow forecasting it doesn't answer the question on operators' minds—will a new forecast tool improve my operations enough to justify the investment in infrastructure, staff training and staff time during transition?~~ Operational evaluations can fit more readily into existing practice than retrospective studies (Krajewski et al. 2021). However, a common ranking does not answer whether a product improves decisions at a specific utility enough to cover deployment, training, and transition costs.

~~Here we present an approach for assisting local water managers and hydropower utility operators in performing a context-specific evaluation of streamflow forecasting and hydropower scheduling tools. The approach calls for broad integration of the operator including~~ We present an operating procedure for this setting. Operators participate in the experimental design, metric selection, tailored metric design, and comparative analysis of the results. We then demonstrate this approach in the context of three hydropower plants operated by select and revise metrics, and interpret results against recent decisions. The evaluation uses the data available in daily operations. These features distinguish the procedure from a fixed retrospective verification.

We demonstrate the procedure at three Great River Hydro, ~~the largest conventional hydropower provider in the Northeastern U. S.~~ By using multiple traditional and tailored metrics as well as making the data easily accessible and comparable, this approach provides operators with the information they need to make informed decisions on forecast products with confidence. Importantly, this approach is designed to be easily replicated on other river systems plants on the Connecticut River. The case study combines standard metrics with measures tied to day-ahead bidding and forebay compliance. The same process can be adapted to other systems, but the tailored metrics will differ.

2 | METHODS

While probabilistic and ensemble forecasts and evaluation metrics are gaining in popularity, deterministic forecasts such as those issued by the National Weather Service River Forecast Centers (RFC) are still widely used operationally and are adopted as a ~~valued benchmark of~~ benchmark in this framework. All three products evaluated here are deterministic, so the metrics we describe are deterministic; Section 2.5 sets out how the framework extends to ensemble products, since the structure of the process does not depend on which kind of forecast is being evaluated. In addition, we focus on metrics which are feasible

[†] www.hepex.org

[‡] <https://www.usbr.gov/research/challenges/forecastrodeo.html>

to compute in a ~~real-time~~ real-time setting as opposed to a hindcast based evaluation. ~~While a hindcast evaluation may be preferable statistically, hindcast evaluations require that forecasters predict (manually or with an operations model) their own behavior during hypothetical past conditions where other factors such as market prices~~

Hindcasts remain the strongest basis for evaluating forecast quality because they provide large samples and a broad range of hydrologic conditions. Comparing hindcast inflow with observed or calculated inflow does not require reconstructing operations. Reconstructing forecasts that include human judgment is harder because the forecaster would need the information available at each historical issue time. Historical day-ahead and ~~data curation issues~~ real-time wholesale prices can often be retrieved, but outages, constraints, operator judgment, and the way price signals affected decisions may not be known. In the remainder of this section we describe the components of a real time forecast evaluation framework for reservoir inflow forecasts including visualizations to convey the evaluation results and an iterative feedback process from the operators recoverable. Extending the analysis from quality to value also requires a reservoir operations model or operator input. A real-time evaluation builds the forecast archive while these conditions occur.

2.1 | ~~real-time~~ Real-time inflow forecast evaluation ~~A general real-time~~

A general real-time forecast evaluation process can be described in the following steps: Steps 1 through 3 are generic practice, common to any verification exercise. Steps 4 through 6 are where this framework departs from retrospective verification.

1. **Collect ~~real-time~~ real-time data** - This may include streamflow forecasts (natural or regulated), calculated inflows, scheduled and actual outflows, and other data relevant for computing evaluation-relevant metrics such as forecast issue times and past forecasts.
2. **Compute evaluation metrics** - At the start of an evaluation, it may only be possible to compute a limited number of generic evaluation metrics but as the available data grows it may become possible to compute more reliable metrics as well as metrics tailored to the specific system.
3. **~~Visualization~~ Visualize results** - ~~Developing easy-to-interpret visuals is critical to the success of an evaluation. Visuals should be concise, clear, and require a limited amount of explanation, or come with built-in documentation~~ Use figures that allow operators to compare products without extensive explanation.
4. **~~Solicit~~ Gather feedback** - Collecting feedback on an ongoing basis from operators is critical to the success of a ~~real-time~~ real-time forecast evaluation. ~~This process can be used~~ Operators should be asked which recent forecast errors mattered operationally and which did not, since the answers are not always apparent from the metrics alone.

5. **Update metrics** - Feedback is used to add metrics tailored to ~~develop tailored metrics that are specific to~~ the system or reservoir being evaluated, ~~and to remove or modify metrics that have proven uninformative.~~ The set of metrics is therefore not fixed at the outset but develops as the evaluation proceeds.
6. **Iterate** - The above steps should be repeated at a regular interval (monthly or weekly) throughout the entire forecast evaluation period, ideally at least one year in duration.

The ~~iterative nature of this evaluation process with direct operator feedback is important to its success. As more data becomes available so does an understanding of the~~ mechanics of steps 4 and 5 determine what the evaluation produces, so an example of how feedback was collected and acted on in our case helps clarify this point. Feedback was collected in recurring meetings with the operators, held monthly for most of the evaluation. Each meeting reviewed the current calculated metrics against recent operating experience, and operators were asked to identify which recent forecast errors mattered operationally, and which did not. The answers frequently diverged from what the generic metrics indicated, especially during the high-flow season. Operators discounted large errors during high-flow spill conditions, when they had little discretion over releases and forecast quality was operationally irrelevant, while treating moderate errors with the potential to impact the forebay compliance band as serious. These ongoing discussions motivated the tailored metrics in Section 3.6: the forebay exceedance metrics were designed in direct response to operators stating that compliance risk, rather than average error, determined whether they trusted a forecast ~~performance which may help to develop additional tailored metrics. The following sections detail some additional considerations.~~

Metric development and data collection occur on different timescales. In this case the metric set stabilized within about six months, while the evaluation continued to sample more conditions. For applications with strong seasonality, including this one, we recommend at least one annual cycle because product rankings can vary by season (Section 3.5). A shorter evaluation remains conditional on the seasons and events observed.

2.2 | Benchmark forecasts

~~It is critical to include benchmark forecasts in the evaluation to provide forecast evaluators with context for how well a specific forecast performs over the benchmarks. Two commonly used benchmarks are the~~ Benchmark forecasts put product performance in context. A “perfect” ~~forecasts, which use forecast substitutes~~ observed or calculated inflows in place of a forecast to get a sense of the best possible forecast, and “persistence” forecasts which use the last observed value carried out for all future timesteps to get a sense of the arguably worst possible forecast. Persistence is a competitive benchmark for short term operations, ~~inflow~~ for a forecast and provides an upper reference. A persistence forecast carries the most recent observation through the forecast horizon. Persistence can be competitive from hours to 2-day horizons. ~~We note that other simple climatologically driven~~

forecasts could be used in place of the persistence forecast when looking at longer horizons like monthly and seasons. Other benchmarks could include publicly available forecast products or any other forecast that is currently being about two days, but it degrades quickly. Climatology is usually a more suitable lower reference at weekly to seasonal horizons. The benchmark should match the decision horizon and may also include the forecast currently used at the reservoir.

NSE and KGE also depend on the distribution of observations at each location, so raw values require care in cross-site comparisons (Knoben et al. 2019). NSE measures improvement over the observed mean. A forecast can therefore have a higher NSE at a more variable site even when its absolute errors are larger. We report RMSE and reference forecasts alongside NSE and KGE so readers can see both absolute error and performance relative to simple alternatives.

2.3 | Smoothing calculated inflows

Most available reservoir inflow data is not directly measured, instead it is computed based on measured Reservoir inflow is often calculated from changes in forebay elevation and a known an elevation-volume relationship. These calculations are often noisy due to short term fluctuations in forebay elevations from rather than measured directly. Short-term changes caused by wind, turbine and spillway release changes, and other such as reservoir seiches or waves propagating from upstream. Inflow computed in this way can often contain negative values and other non-physical fluctuations. In this situation we recommend the inflow data be smoothed prior to comparing with any forecasted releases, seiches, and upstream waves can produce noise and nonphysical values. Smoothing may therefore be appropriate before comparing calculated and forecast inflow. We recommend a triangular smoother that applies moving averages using a triangular weight function. This is a simple option which produces reasonable results but many more options used two successive moving averages, which produce a triangular kernel, but other filters are available (Elshorbagy et al. 2002). Figure 3 shows an example of hourly computed inflow data which has been smoothed using a triangular smoother. The window must be reported and tested because smoothing changes the evaluation target.

The top figure shows the unsmoothed inflow and bottom figure shows the smoothed inflow used in the evaluation. The black lines show the calculated inflow based on forebay observations and dam outflow and the gray lines show the forecasts for reference.

2.4 | Traditional inflow evaluation metrics

Traditional deterministic forecast evaluation metrics that are particularly relevant to inflow forecasting include Deterministic inflow forecasts can be evaluated using measures of average behavior error, correlation, and bias. Additional metrics include higher order moments of the forecast distributions, categorical forecast. Higher moments, categorical performance, and conditional statistics (St-Aubin and Agard 2022 and Hewamalage et al. 2023). This section is not intended to be a

~~comprehensive review of deterministic forecast evaluation metrics~~. Therefore, we recommend some simple metrics which can be augmented as needed for particular applications. These metrics should be computed for a variety of lead times as well as seasonally to fully assess forecast performance across a range of conditions may also be useful (St-Aubin and Agard 2022 and Hewamalage et al. 2023). We use a small set of metrics that can be expanded for a particular application. Each should be calculated by lead time and season because aggregate values can hide changes in performance.

Average hydrologic forecast behavior is typically measured with an error metric such as root mean squared error (RMSE), Nash-Sutcliffe Efficiency (NSE) ~~, or King-Gupta~~ (Nash and Sutcliffe 1970), or Kling-Gupta Efficiency (KGE) ~~(refs)~~ (Gupta et al. 2009). We recommend KGE because it ~~is not skewed by large values where~~ decomposes forecast performance into correlation, a variability ratio, and a bias ratio, so that a low score can be attributed to a specific deficiency rather than only registered as large aggregate error. NSE and RMSE ~~are due to the use of a squared error term~~ aggregate squared error into a single number, which makes them more strongly influenced by the largest errors and less diagnostic about their source. The KGE is given by

$$KGE = 1 - \sqrt{(r - 1)^2 + \left(\frac{\sigma_f}{\sigma_o} - 1\right)^2 + \left(\frac{\mu_f}{\mu_o} - 1\right)^2} \quad (1)$$

where r is the linear correlation coefficient, σ_f ~~is the standard deviation of the forecast, and~~ σ_o ~~is the standard deviation of the observations,~~ are forecast and observed standard deviations, and μ_f ~~is the mean of the forecast, and~~ μ_o ~~is the mean of the observations.~~ Values of KGE that are greater than -0.41 are better than the mean of the observations, but larger values are preferred up to the maximum value of 1 for a perfect forecast. We note that are forecast and observed means. KGE values greater than -0.41 improve on the observed mean as a predictor (Knoben et al. 2019); the maximum is 1. NSE and KGE have different properties and ~~so~~ should not be used interchangeably ~~(Knoben et al. 2019)~~. We report both and use RMSE, bias, correlation, and benchmark forecasts to interpret them.

The bias error, or residual, of a single forecast ~~point~~ is the difference between ~~observed and forecasted values but this may be expressed as a percentage~~ the forecast and the observation. Bias is the systematic component of error, the tendency of a forecast to be too high or too low on average, and is conventionally expressed for a set of forecasts ~~by averaging across all bias values as a percentage of total observed volume~~

$$PBIAS = 100 [\%] \times \frac{\sum (f_i - o_i)}{\sum (o_i)} \quad (2)$$

where ~~PBIAS measures the tendency of a forecast (f) to be greater than (positive values) or less than (negative values) than observations~~ (o) positive values indicate forecasts greater than observations and negative values indicate forecasts less than observations.

Correlation measures the strength of a linear relationship. Linear association between observations and forecasts is computed measured using the Pearson correlation coefficient

$$r = \frac{\sum (o_i - \mu_o)(f_i - \mu_f)}{\sqrt{\sum (o_i - \mu_o)^2} \sqrt{\sum (f_i - \mu_f)^2}} \quad (3)$$

where r can range from -1 to 1 but for forecast evaluation, negative values would indicate poor forecast performance.

Metrics should be computed on a common sample when products cover different periods. Skill scores relative to a reference forecast are also useful, but the evaluated and reference forecasts must use the same observations. Otherwise differences in sample variance can be mistaken for differences in skill.

2.5 | Extension to probabilistic and ensemble forecasts

The process applies to deterministic and probabilistic forecasts; the metrics and figures change. Probabilistic products can describe risk around compliance or market decisions. The continuous ranked probability score (CRPS), rank histograms, and reliability diagrams are common choices. CRPS reduces to mean absolute error for a deterministic forecast, which permits comparison on a common scale.

3 | CASE STUDY: GREAT RIVER HYDRO

Great River Hydro (GRH) is New England's largest conventional hydropower generator. As a private, independent power producer, GRH owns and operates 589 megawatts of nominal hydropower capacity installed nameplate hydropower capacity across 13 generating stations on the Connecticut and Deerfield Rivers in ISO New England. In this evaluation we focused on the inflow to three main hydropower producing dams in the system, Wilder, Bellows Falls, and Vernon (Figure 1) as well as three main tributaries, the Ottauquechee, Sugar and White Rivers.

The three dams lie in series on the main stem of the Connecticut River, with Wilder upstream, followed by Bellows Falls, then Vernon. The three projects are operated jointly, so releases from each become inflow to the next. The only significant local flows occur between Wilder and Bellows Falls in the form of the three tributaries mentioned above. These three sites were selected from the larger set shown in Figure 1 because they are the largest generators on the system and they are where the operators concentrate their scheduling attention. In addition, these projects are part of an ongoing relicensing process through the Federal Energy Regulatory Commission (FERC), which includes new limits to forebay fluctuations. The remaining facilities in Figure 1 are smaller and are not covered by the same license conditions.

Basin hydrology is snowmelt influenced, with a spring freshet typically spanning March through May, convective rainfall in summer, and rain and rain-on-snow driven events in winter. Several tributaries have upstream flood control regulation by the U.S. Army Corps of Engineers, which contributes to inflow variability that is not directly weather driven.

TABLE 1 Physical characteristics of the six evaluation points. Drainage areas are from the USGS National Water Information System for the reference gage listed. Mainstem areas are nested, so the area above Vernon includes those above Bellows Falls and Wilder.

Location	Type	USGS reference gage	Area (mi ²)	Area (km ²)
Wilder	Mainstem dam	01144500 West Lebanon, NH	4,092	10,598
Bellows Falls	Mainstem dam	01154500 North Walpole, NH	5,493	14,227
Vernon	Mainstem dam	01156500 Vernon, VT	6,266	16,229
White River	Tributary	01144000 West Hartford, VT	690	1,787
Sugar River	Tributary	01152500 West Claremont, NH	269	697
Ottawaquechee River	Tributary	01151500 North Hartland, VT	221	572

3.1 | Operating context

The three dams have limited storage relative to their inflow and are operated as run-of-river facilities. Their daily cycle consists of an upward ramp, a period of flexible generation, a downward ramp, and a refill period. Generation is bid into the ISO New England day-ahead market, so the day-ahead inflow volume is the quantity that matters most to the operators, since it sets what they can bid into the market. The binding operational constraint comes from the FERC relicensing of the three lower Connecticut River dams, which imposed a ± 0.5 foot (≈ 15 cm) limit on forebay fluctuation, with penalties for exceedance. The central operating problem is therefore committing generation based on an uncertain inflow forecast without violating the forebay limits. This is what motivated the cumulative-error and threshold-exceedance metrics in Section 3.6.

At the time of the evaluation the operators used an in-house forecast built from National Weather Service River Forecast Center forecasts, routed to the dam locations and adjusted using operator judgment. This is the incumbent product against which the two commercial forecasts were evaluated, and it is labeled GRH in the results.

Setting up a **real-time-real-time** evaluation required the development of an automated data platform that enabled sharing of **real-time-real-time** system data as well as the forecasts. This was necessary to share certain data with external parties for the purposes of forecast evaluation. While such a system may not be necessary for evaluations conducted internally by a single organization, some form of data access (API, database, etc.) is required for a **real-time-real-time** forecast evaluation. Such systems incur an up-front cost that should be considered when deciding on an evaluation approach. In addition, **real-time-real-time** evaluations come with all of the challenges of any operational system, such as system uptime, latency, and maintenance. **These factors may help decide if a forecast is viable before any evaluation is done; a forecast that cannot be produced operationally should not be considered in a real-time evaluation.**

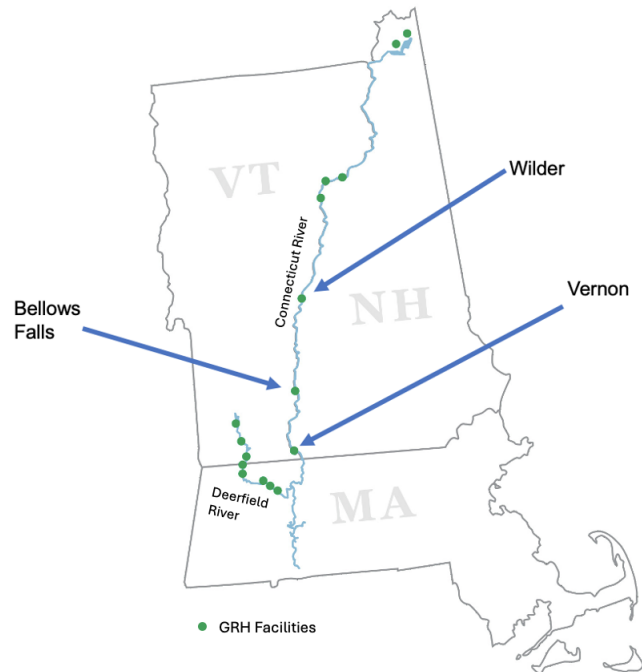


FIGURE 1 The Great River Hydro system in the Northeast U.S. with the locations of the three major dams in this evaluation.

We ran the ~~real-time~~ real-time evaluation for almost ~~2-years~~ from July two years, from June 2023 to ~~May~~ April 2025, with regular feedback from the operators. We evaluated two commercial forecast products which are anonymized and labeled as A and B. In addition we evaluated the GRH in-house forecast which is a combination of RFC forecasts, routing, and operator judgment. During the initial few months of the evaluation we focused on generic metrics of forecast performance. As the quantity of data increased we transitioned to seasonally computed metrics and tailored metrics developed in collaboration with the system operators.

~~While most weather-driven inflow forecasts are available hourly for the next 7-10 days, The three products differ in the coverage they provided over this period, which constrains how they can be compared. Forecast B was available for the volumetric forecast for the next day is the most important for GRH. This largely sets what they can bid into the day ahead market. Table 2) show the overall forecast evaluation metrics for the two commercial forecasts (A full period, giving 657 day-ahead forecasts at each dam. Forecast A was discontinued in November 2024 and the in-house forecast archive also ends in November 2024, giving 539 and 528 day-ahead forecasts respectively. The three products overlap on 510 days at each dam, from 3 June 2023 to 20 November 2024. Metrics computed over each product's full available record, which is what we report, are therefore not computed over identical samples. We repeated the calculation restricted to the 510 common days and the ranking of the three products is unchanged at all three dams, with individual metrics shifting by no more than 0.03 NSE, so we report the full-record values so as not to discard any data. This kind of partial coverage reflects the operational realities of any real-time evaluation.~~

3.2 | Forecast products

Anonymization of the commercial products limits reproducibility, so we describe each to the extent our agreements allow. Forecast A is produced by a consulting firm using a conceptual rainfall-runoff model with hydraulic routing through the Connecticut River main stem, driven by a commercial numerical weather prediction forecast, and issued once daily at an hourly time step out to roughly 8 days. Forecast B is produced by a technology firm using a machine learning model trained on gauge and remotely sensed catchment observations, issued twice daily at an hourly time step out to 10 days. The GRH in-house forecast combines RFC forecasts with routing and operator judgment and extends only through the day-ahead period, roughly 32 hours, because that is the horizon it is used for. Neither vendor released model internals or calibration and validation diagnostics, and we did not have access to them. A utility evaluating commercial forecast products generally cannot inspect them and must judge them on operational performance, which is part of the motivation for this framework. A full procurement evaluation should also weigh the operational cost, data dependencies, and ease of integration of each product alongside its skill, since these differ substantially between the two commercial products evaluated here.

3.3 | Hydrology of the evaluation period

The evaluation period included the July 2023 Vermont flood. This event produced the largest calculated inflows of the evaluation at Vernon (83, ~~B~~ 100 cfs, 2, ~~the GRH~~ 353 m³/s) and Bellows Falls (67,000 cfs). A December 2023 rain-on-snow event produced the largest inflow at Wilder (27,600 cfs) and the second largest at Vernon and Bellows Falls. The evaluation period also included spring freshets of 2024 and 2025 and dry summer conditions in both 2023 and 2024. The period therefore sampled a reasonable range of the conditions relevant to GRH operations. It did not, however, sample the full distribution of hydrologic conditions. Notably, the July 2023 event produced one of the ten largest flows ever recorded on the main stem of the Connecticut River. The evaluation also did not include any severe multi-year drought. Any real-time evaluation is bounded in this way, and the conclusions below should be taken as conditional on the observed period.

Most weather-driven inflow forecasts extend 7–10 days, but GRH relies most on next-day volume because it informs the day-ahead market bid. Table 2 compares the two commercial products, the in-house forecast (GRH), and the benchmarks-perfect and persistence forecasts for the one-day-ahead forecasts. Overall we found that for this time period, GRH's benchmarks for this horizon. Over the evaluation period, forecast B and the in-house forecast was roughly similar to forecast B but better than forecast A overall. This overall tabular view is a good way to get quick comparisons between forecasts but can obfuscate certain forecast behaviors, especially seasonal. performed similarly and both outperformed A. Lead-time and monthly results below show variation hidden by the aggregate table.

Table 2 also shows why cross-location comparisons need more than one metric. Forecast A has its highest NSE at Vernon (0.81) and its lowest at Bellows Falls (0.55), yet its RMSE is 5,097 cfs at Vernon, 5,904 cfs at Bellows Falls, and 2,278 cfs at Wilder. NSE is normalized by observed variance, so a higher value at one site does not imply a smaller error than at another. The persistence benchmark also changes sharply among sites, from an NSE of -0.81 at Wilder to 0.80 at Vernon. We therefore use the table to compare products within a location, while RMSE and the benchmark rows provide context across locations. Metrics computed on the 510 days common to all products preserve the within-location ranking.

TABLE 2 Overall Day-ahead forecast evaluation metrics for the real-time evaluation conducted from July–June 2023 to May 2025 for the two commercial forecasts (April 2025, A, and B) are commercial products, the GRH is the in-house forecast (GRH), and the benchmarks perfect and persistence are benchmarks. The products cover different portions of the period; n is the number of forecasts evaluated. Values should be compared primarily within a location because NSE and KGE depend on the local observation distribution.

Location	Forecast	Location- n	NSE	KGE	PBIAS (%)	RMSE (cfs)	r
Wilder	A	Wilder-539	0.72	0.83	-8.00 – -8.0	2,278	0.88
	B	Wilder-657	0.83	0.88	0.10 – 0.1	1,786	0.93
	GRH	Wilder-528	0.88	0.89	-9.90 – -9.9	1,476	0.95
	perfect	Wilder-657	1.00	1.00	0.00 – 0.0	0	1.00
	persistence	Wilder-657	-0.81 – -0.81	0.02	-75.90 – -75.9	5,866	0.75
Bellows Falls	A	Bellows-539	0.55	0.77	0.40 – 0.4	5,904	0.77
	B	Bellows-657	0.91	0.95	0.60 – 0.6	2,615	0.95
	GRH	Bellows-528	0.85	0.88	-10.40 – -10.4	3,309	0.93
	perfect	Bellows-657	1.00	1.00	0.00 – 0.0	0	1.00
	persistence	Bellows-657	0.39	0.35	-39.20 – -39.2	6,831	0.87
Vernon	A	Vernon-539	0.81	0.83	-11.30 – -11.3	5,097	0.91
	B	Vernon-657	0.89	0.94	-1.70 – -1.7	3,872	0.94
	GRH	Vernon-528	0.89	0.88	-11.00 – -11.0	3,764	0.95
	perfect	Vernon-657	1.00	1.00	0.00 – 0.0	0	1.00
	persistence	Vernon-657	0.80	0.69	-18.10 – -18.1	5,167	0.93

Figure 2 shows the performance of all the available forecasts and benchmarks out to a 10 day lead time. The GRH in-house forecast is only used to develop bids for the day-ahead-day-ahead market so it does not extend the full 10 day period. At Wilder and Vernon all three forecasts perform roughly similar with 1–3 days of lead time with the A and B forecasts diverging after that period. At Bellows Falls the three forecasts are more distinct in terms of quality with the B showing the best performance at most lead times followed by the GRH in-house forecast and then forecast A. Notably, after three days of Both commercial forecasts develop a growing negative bias with lead time, all forecasts exhibited some amount of bias at the downstream forecast locations which may be due to errors propagating from upstream and the bias grows further downstream, consistent with errors propagating through the cascading reservoirs. For forecast A, day-ahead PBIAS is 0.4% at Bellows Falls and grows to -19.1% by day 7; for forecast B the corresponding values are -1.3% and -7.8% . Because releases from each dam become inflow to the next, an error propagates directly to the next downstream project, and the effect compounds with lead time.

Figure 4 shows metrics by month for the entire evaluation period which reveals some trends in forecast quality in certain seasons. Namely forecasts tend to perform worse in the winter which is the wet season and worse in the summer which is the

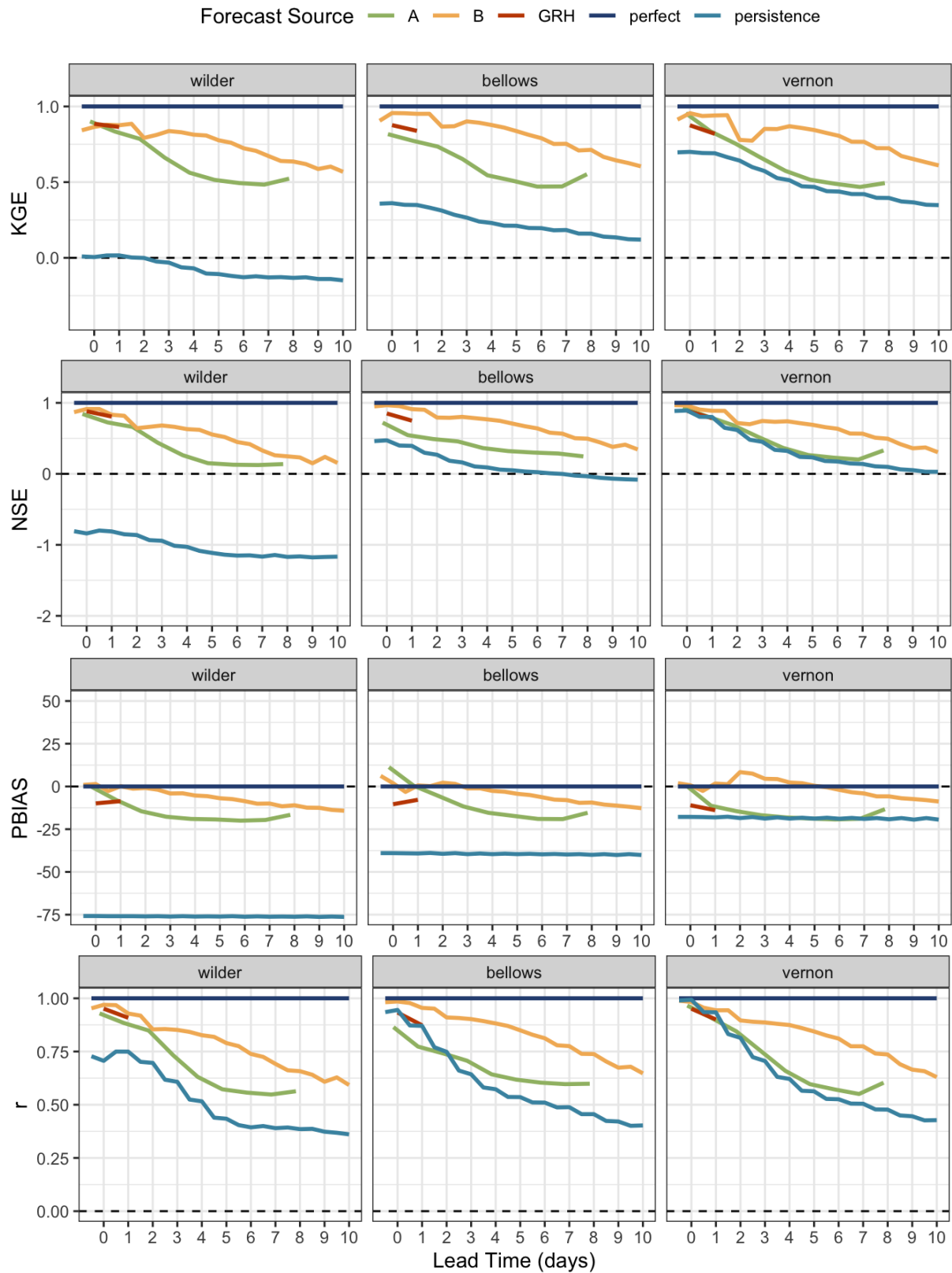


FIGURE 2 Forecast performance metrics by lead time. A and B are commercial forecasts, GRH is an in-house forecast. The dashed line indicates the threshold where a forecast has no skill over the mean of the observations, which is NSE of 0 and KGE of -0.41. The persistence benchmark is a meaningful reference only over the first two days; beyond that it is shown for reference purposes only.

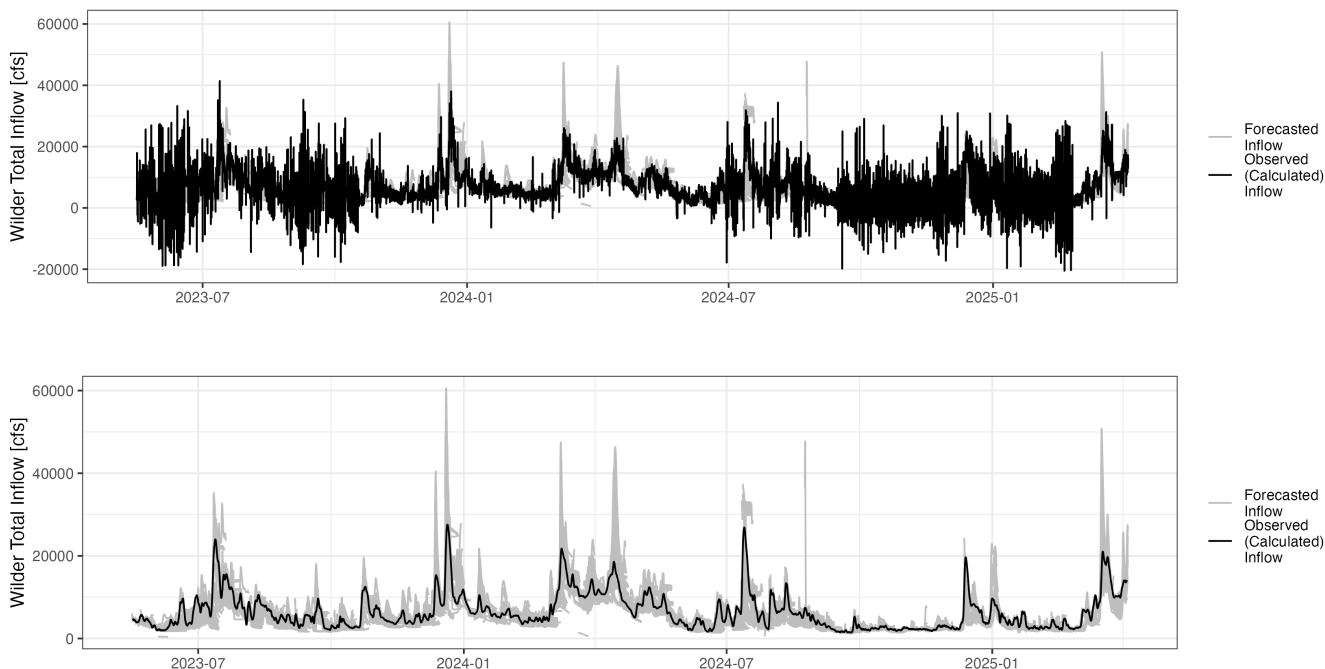


FIGURE 3 Unsmoothed (top) and smoothed (bottom) calculated inflow at Wilder Dam. Black lines show calculated inflow derived from forebay observations and dam outflow; gray lines show forecasts. Smoothing uses two successive 24-hour moving averages, yielding a triangular kernel spanning 47 hourly observations. Flows are in cubic feet per second (1,000 cfs $\approx$ 28.3 m³/s).

dry-season. Forecasts tended to perform well during the snow-melt season (April-July). The natural hydrology of the system is driven by different mechanisms in different seasons, for example, snow vs rain and baseflow vs direct runoff, which may explain the varying model performance by season. Although forecast B tends to perform better overall, there are cases in which the A-forecast is better, highlighting the difficulty of selecting a single deterministic forecast.

3.4 | Smoothing GRH calculated inflows

Figure 3 shows calculated inflow before and after smoothing. We applied two successive 24-hour moving averages. Their convolution yields a triangular kernel spanning 47 hourly observations; “24-hour” refers to the width of each component moving average. We selected this width to match the day-ahead volumetric decision. We repeated the evaluation with component windows of 0, 6, 12, and 24 hours (Table 3). The largest NSE change was 0.05, and the product ranking was unchanged at every location. Forecast B led A at Bellows Falls by 0.39 NSE without smoothing and by 0.37 with the 24-hour component window. Evaluators should report the filter and test the sensitivity of conclusions to its width.

TABLE 3 Sensitivity of day-ahead NSE to the component moving-average width used to smooth calculated inflow. Each nonzero width is applied twice, producing a triangular kernel. The 24-hour column corresponds to Table 2. Product rankings are unchanged.

Location	Forecast	0 h	6 h	12 h	24 h
Wilder	A	0.70	0.72	0.72	0.72
	B	0.82	0.83	0.83	0.83
	GRH	0.87	0.87	0.88	0.88
Bellows Falls	A	0.50	0.51	0.52	0.55
	B	0.90	0.90	0.90	0.91
	GRH	0.84	0.83	0.84	0.85
Vernon	A	0.79	0.79	0.79	0.81
	B	0.88	0.88	0.88	0.89
	GRH	0.88	0.88	0.88	0.89

3.5 | Seasonal performance

Figure 4 shows systematic monthly variation in forecast quality. Skill is highest during the spring snowmelt period, roughly March through May, for all three products. It is lowest in late summer and fall for the commercial products. Forecast A also degrades sharply in January and February, when day-ahead KGE pooled across the three dams falls to 0.45 and -0.67 , respectively; the February value is below the plotted range. Performance is poorest when inflow depends on rain, rain-on-snow, or convective storms. These events depend on uncertain precipitation forecasts. Snowmelt inflow is more predictable because the snowpack can be observed before it melts and temperature forecasts are more reliable than precipitation forecasts. Forecast A outperforms B in April and May, so no product leads in every month.

3.6 | Tailored metrics

In addition to the generic forecast evaluation metrics described above, we developed several tailored metrics specifically relevant to the GRH system and its operations. Each arose from the feedback process described in steps 4 and 5 of Section 2.1, and Table 4 traces each metric to the operator concern that prompted it. The first focused on day-ahead-day-ahead forecast performance as shown in the previous section. This was motivated by the day-ahead-day-ahead market that GRH bids into, as such they focus their attention heavily on the 1-2 day ahead period where the forecasts matter the most. Operationally, they adjust their flows after this point daily, so no forecast is used after the 2 day mark.

~~During the Federal Energy Regulatory Commission (FERC) relicensing for the three main dams on the lower Connecticut river a strict limit on forebay fluctuations was imposed. GRH was given a +/-~~

~~The FERC relicensing described in Section 3.1 imposed a ± 0.5 foot operating range after which penalties are incurred. Given GRH's reliance on the inflow forecast, a metric was developed that captures the chance of the forebay limit being exceeded if the inflow forecast is used directly (≈ 15 cm) forebay operating band. We calculated cumulative hourly forecast error within each forecast issue and converted the accumulated volume to an equivalent forebay elevation change using an~~

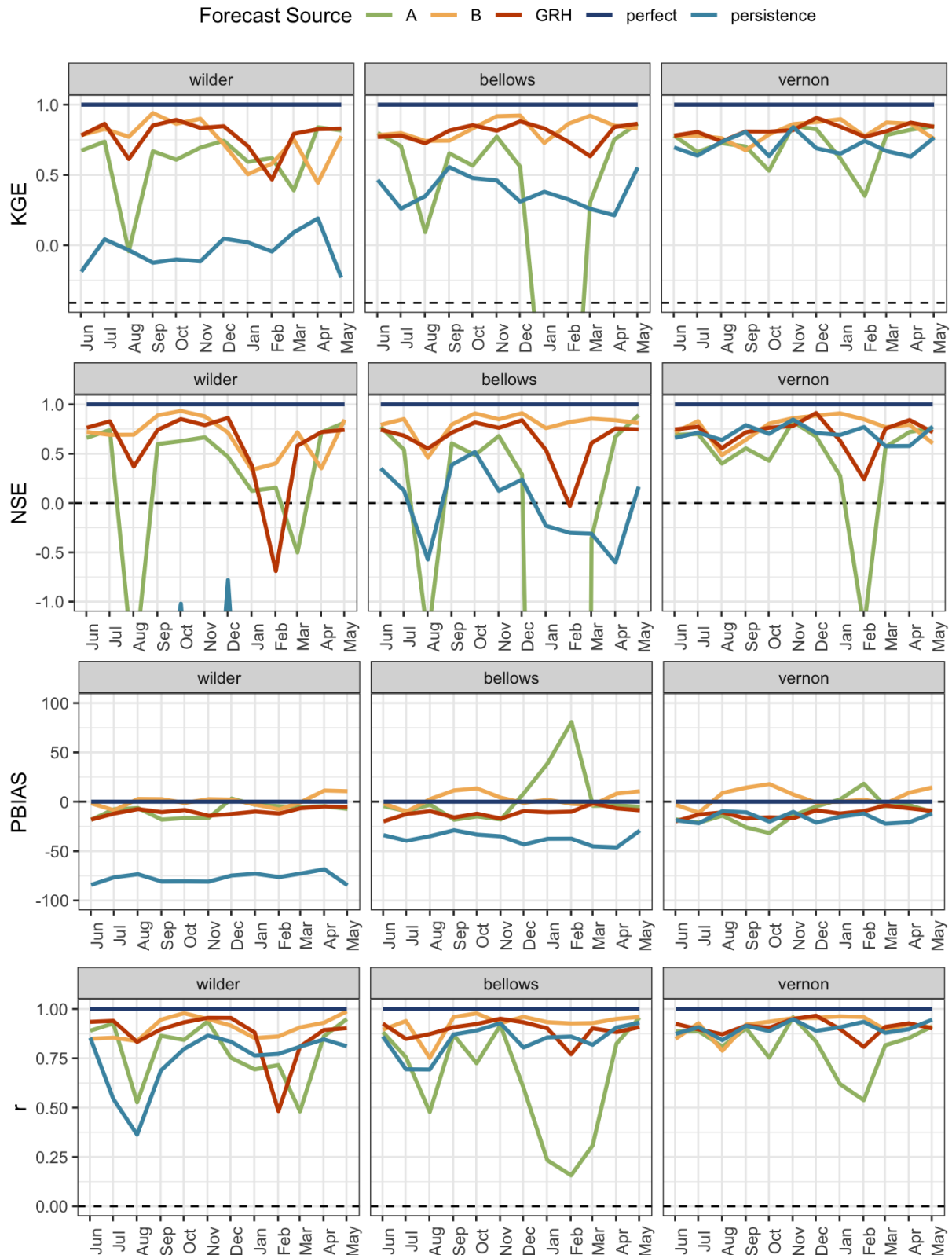


FIGURE 4 Forecast performance metrics by month. A and B are commercial forecasts, GRH is an in-house forecast. The dashed line indicates the threshold where a forecast has no skill over the mean of the observations, which is NSE of 0 and KGE of -0.41.

TABLE 4 Tailored metrics developed during the evaluation, the operator concern that prompted each, and the approximate point in the evaluation at which it was added. Metric development converged within roughly the first six months.

Metric	Operator concern	Added
Day-ahead volumetric skill	Day-ahead market bids are set from next-day inflow volume, not hourly shape	Months 1–2
Seasonal metric breakdown	Products were suspected to behave differently in snowmelt and rain driven regimes	Months 3–4
Cumulative forebay error	Average error does not indicate whether the forebay band would be breached	Months 4–6
Forebay exceedance probability	Operators needed to know how quickly cumulative forecast error could exhaust the forebay band	Months 5–6

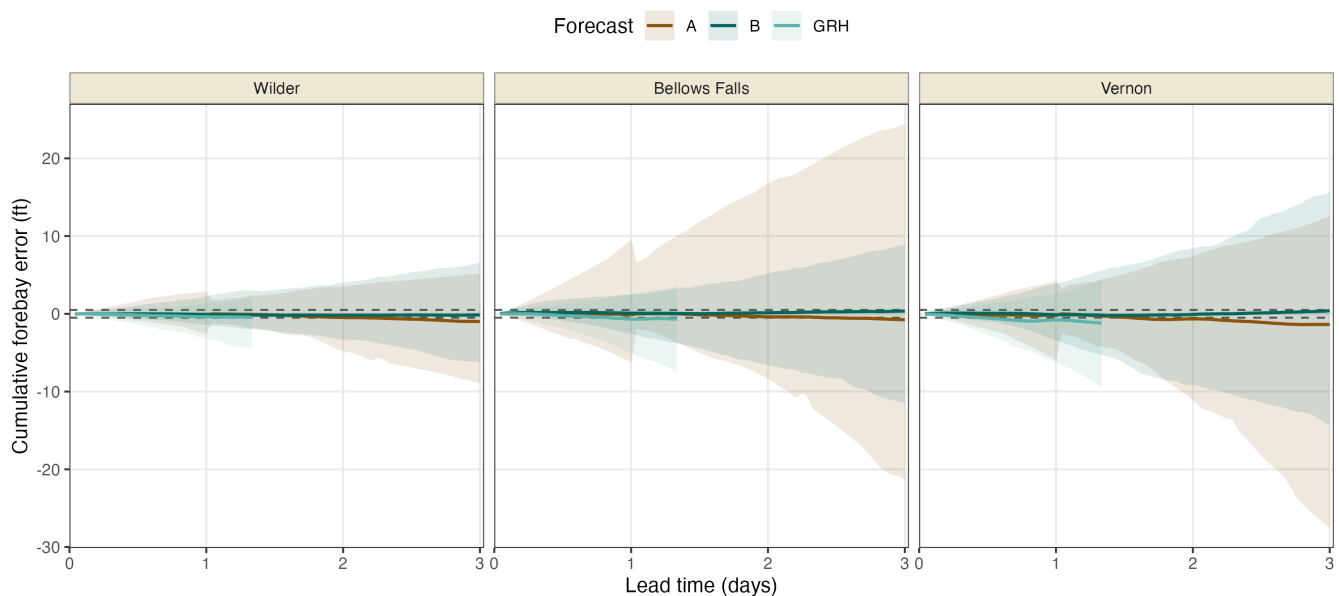


FIGURE 5 The cumulative probability of exceeding Cumulative forebay error under an idealized pass-inflow stress test with a 0.5-foot operating range if neutral starting position. Lines show the operations were set median across forecast issues and shaded bands span the 2.5th to pass-inflow-97.5th percentiles. Dashed lines mark the ± 0.5 foot (inflow=outflow ≈ 15 cm) where inflow was determined by the various inflow forecasts forebay operating band.

approximate storage scale of 30,000 cfs-hours per foot. Figure 5 shows where the forebay would be after a certain amount of time if the forecast was used directly to set a pass inflow condition (i.e., inflow=outflow). The error bars extend to the 2.5th and 97.5th percentile of all forecasts. We can see that rarely would a forecast cause the forebay to exceed the 0.5-foot limit during the first day of operations, after which the operators can adjust the median and middle 95% of the resulting trajectories. This is an idealized stress test: it assumes forecast inflow is passed directly as outflow from a neutral forebay position. GRH does not operate this way. Operators schedule generation to market bids, depart from inflow within the day, and adjust as new information arrives. The metric is therefore neither a prediction nor a bound on observed compliance rates. Estimating realized risk requires a scheduling model and operator review.

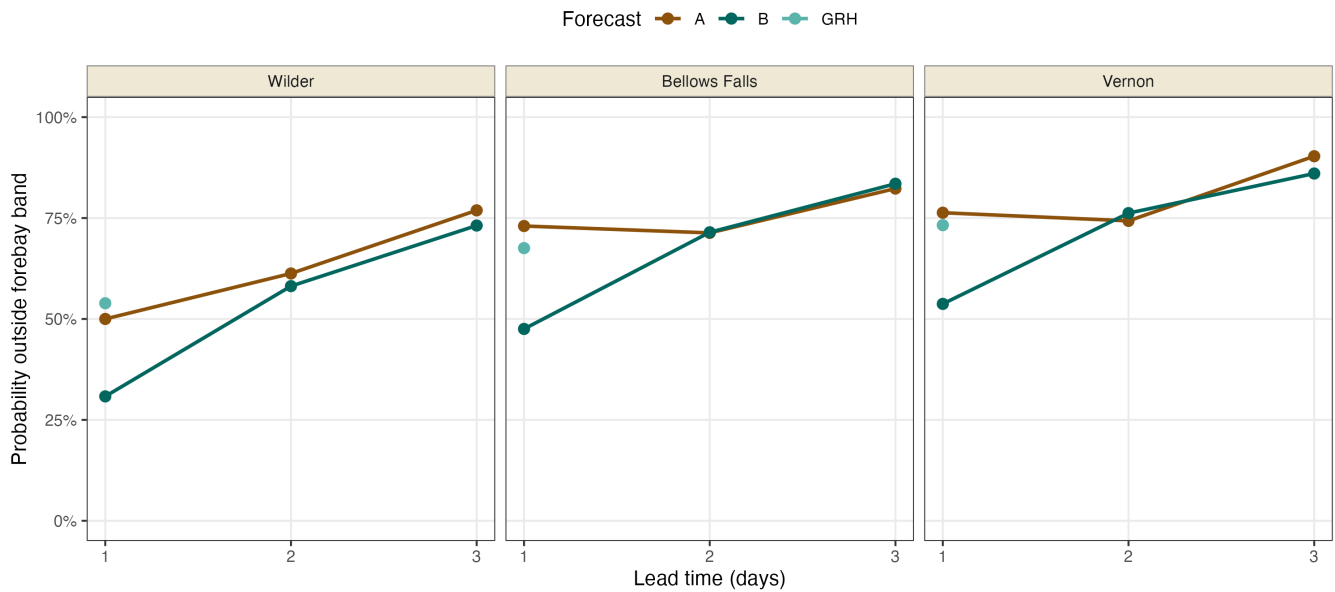


FIGURE 6 Probability of exceeding a the ± 0.5 foot operating range given a neutral forebay starting position on day-0 (top ≈ 15 cm) and probability of exceeding a 0.5 foot forebay operating range on band at daily lead times under the second day, given the idealized pass-inflow stress test. Each forecast issue begins from a neutral forebay starting position from. The in-house forecast ends after the first day (bottom) day-ahead period.

Figure 6 (top) shows the corresponding probability of exceeding the 0.5 foot limit given a neutral forebay starting position on day-0. Results suggest that chances of exceedance are less than 10% for all forecasts for all lead times in the 1-3 day range. GRH was also interested in seeing a more realistic situation involving a non-optimal forebay starting position on the second day, what was the probability of exceeding the 0.5 foot range, which is shown in Figure 6 (bottom). We can see that the probabilities are much higher, up to 25% in some cases for the A forecast. This information provides a detailed view of the forecast performance that GRH can use to assess forecast quality, leaving the forebay band. During the first 24 hours, forecast B has lower exceedance probability than A at all three dams: 30.8% versus 50.0% at Wilder, 47.5% versus 73.0% at Bellows Falls, and 53.7% versus 76.3% at Vernon. The in-house forecast is available only through the day-ahead period and has probabilities of 53.9%, 67.6%, and 73.2%, respectively. By 48 hours, the two commercial products converge to 58.1–76.2% across the three dams; by 72 hours, probabilities range from 73.1% to 90.3%. Thus B provides a clearer advantage over the first day, while neither commercial product can be passed through without intervention over longer horizons. This distinction was not apparent from the aggregate metrics alone.

4 | DISCUSSION

In this paper we present a general framework that can be used in real time (or near real time) for reservoir inflow forecast evaluation. The evaluation can include forecast verification and validation but also include other qualitative or specialized

metrics of forecast quality. The framework calls for an iterative approach which involves direct operator feedback in the process. We also presented a successful application of this evaluation framework to three reservoirs in the Great River Hydro system on the Connecticut River in the Northeast U. S. While there is no single evaluation approach that can apply to all forecasts, we present a range of options of both traditional and tailored metrics that will be suitable for many situations. The goal of such a framework is to provide operators with information necessary for evaluating how much a particular forecast product will improve their operations and ultimately inform ongoing investment decisions (i.e. purchase external forecast products or improve existing internal forecast). We presented a procedure for comparing reservoir inflow forecasts under current operating conditions when hindcasts are unavailable. It combines standard verification with metrics developed from operator concerns. At GRH, product B generally matched the in-house forecast and outperformed A. The forebay stress test identified a day-ahead advantage for B that aggregate metrics did not express in operational terms. The procedure supports product screening, but it does not by itself estimate revenue or compliance outcomes.

In our real-time evaluation of the GRH inflow forecasts there was one commercial forecast that stood out as having better performance overall. However, it was not always the best.

4.1 | Comparison with retrospective forecast evaluation

A retrospective evaluation can provide larger samples, more extremes, and broader hydrologic coverage than the less than two years available here. Forecast quality can be evaluated directly from hindcasts. Reconstructing value is more difficult. Historical wholesale prices may be available, but outages, constraints, operator judgment, and the response to market conditions may not be. A real-time evaluation observes these conditions as they occur, but it cannot replace a hindcast where one is feasible. The two approaches answer complementary questions.

4.2 | From forecast quality to forecast value

The metrics presented here measure forecast quality, not forecast value. Quality describes the relation between forecasts and observations. Value also depends on the decision, its consequences, and the decision maker's risk preferences. A more accurate forecast has no added value when it does not change an action. Classical cost-loss analysis links forecast use to decision utility (Thompson and Brier 1955). The relative utility value framework extends this logic to binary, multicategory, and continuous decisions and can represent risk aversion (Laugesen et al. 2023). Farkas et al. (2025) review related approaches.

The tailored metrics move toward value by expressing error in units tied to a decision. The forebay metric is still a stress test, not a valuation, because it omits actual schedules and operator responses. A GRH value study would first screen products using the quality evaluation, then route the archived forecasts through a scheduling and dispatch model containing reservoir

constraints, unit performance curves, day-ahead and real-time prices, and the incumbent operating strategy. Benefits should be measured relative to the forecast and decisions currently used, with sensitivity tests for risk preferences and alternative operating strategies.

4.3 | Limitations and next steps

Forecast B performed better overall, but it did not lead in every month and every lead time. This underscores the difficulty of selecting a single best deterministic forecast. Moving to a probabilistic forecast may alleviate this somewhat but operators still need to bid in a fixed amount into energy markets so overall improvements in deterministic forecast skill are valuable. In this study we did not address how much improvements in inflow forecasts will translate into improvements in hydropower operations and increased revenue. An important next step in this evaluation framework is to couple this evaluation with hydropower scheduling models to quantify increases in revenue from improved inflow forecasts. While this seems a straight forward next steps, many hydropower scheduling models work with ranges of operations that match the utility's bidding strategy and local market opportunities. This complexity implies that evaluating the opportunities for harnessing higher forecast accuracy may also requires updating scheduling models and exploring more monetized operations or at every horizon. Product selection should therefore consider the decisions and seasons that matter most. Probabilistic forecasts can describe uncertainty, although operators must still submit fixed market bids.

A second lesson concerns comparisons among locations. Raw efficiency metrics depend on observed variability, so they should not be used alone to rank sites. RMSE, sample coverage, and reference forecasts provide necessary context. Comparisons among products should use the same dates whenever coverage differs.

We did not estimate how forecast improvements affect operations or revenue. That analysis would couple the archived forecasts to a hydropower scheduling model. A model configured to reproduce the current bidding strategy may understate the benefit of a more accurate forecast because the strategy was developed around the incumbent product. A value study may therefore need to revise the scheduling model and operating strategy along with the forecast input.

Real-time forecast evaluations are inherently limited by the amount of time they are conducted. Ideally a forecast model's performance would be assessed over the full range of hydrologic conditions spanning multiple decades. A hindcast evaluation may account for the full range of hydrologic variability but this requires an operations model or considerable effort on the part of the system operators to re-operate the system for every hindcast. The decision to pursue a real-time or hindcast evaluation must be made by the system operators depending on their specific system needs and resource availability. The record length limits the real-time evaluation. The period included a regional flood and a rain-on-snow event (Section 3.3), but no multi-year

drought. We could not assess product performance under sustained water-limited conditions. A multi-decade hindcast would cover more hydrologic regimes.

The choice between real-time and retrospective evaluation depends on the system and available resources. We recommend both when feasible. For this seasonally varying system, at least one year of real-time evaluation is appropriate. A shorter period samples only part of the annual cycle, so its conclusions are conditional on the seasons and events observed.

~~The iterative nature of this framework is perhaps its most valuable component. Having direct operator involvement and feedback helps to provide value to the operators. The development of tailored metrics make the results relevant to a particular system and helps with operator buy-in. Furthermore, both generic and tailored metrics can ultimately be extended to assess the impacts of inflow forecasts on hydropower scheduling. Particularly under dynamic electricity market conditions, evaluations of inflow forecasts can serve as key indicators for understanding how forecasting errors propagate through the system, influencing compliance with operational requirements, variations in actual power generation, and potential revenue for system operators. The analysis of inflow forecast performance, conducted in coordination with system operators, represents an important foundational step toward achieving these broader applications. In presenting this framework we hope to provide a template for further applications and unbiased evaluation of the myriad of inflow forecasts available to hydropower system operators.~~ reference inflows are calculated from forebay elevation changes and dam outflow. The observations therefore depend on operations and are not independent in the way a gauge record would be. The three products also cover different portions of the evaluation period (Section 3.1), so Table 2 uses overlapping but nonidentical samples. Restricting the analysis to common dates does not change the ranking. We report the full records to retain the available data.

Operator feedback determines which errors matter and which metrics should be revised or removed. Generic metrics support comparison. Tailored metrics connect errors to site constraints. Together they provide the forecast archive and quality screen needed for a later scheduling and value analysis.

DATA AND CODE

Code for the evaluation is freely available[§]. Sample data can be accessed from the data repository on Zenodo[¶].

Note that a pre-print non-peer reviewed draft of the manuscript has been uploaded to <https://eartharxiv.org/repository/view/10530/>.

AUTHOR CONTRIBUTIONS

Cameron Bracken - Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - Original Draft Preparation. Youngjun Son - Conceptualization, Data Curation, Formal Analysis, Investigation, Software, Validation, Visualization, Writing - Original Draft Preparation. Vince Tidwell - Conceptualization,

[§] <https://github.com/HydroWires-PNNL/inflow-forecast-evaluation>

[¶] <https://doi.org/10.5281/zenodo.16921728>

Formal Analysis, Investigation, Methodology, Project Administration, Supervision, Writing - Original Draft Preparation. Nathalie Voisin - Conceptualization, Methodology, Project Administration, Supervision, Funding ~~Acquisition, Writing~~—Acquisition, Writing - Original Draft.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

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